Human-Like Next-Link Prediction on Wikipedia: A Hybrid GNN Approach

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Abstract

Humans navigating Wikipedia do not strictly follow shortest paths; they rely on semantic intuition, visual cues, and structural context. This project aims to model this human navigation strategy by predicting the next link a player will click in the *Wikispeedia* game. We propose a hybrid architecture that combines semantic understanding (via Sentence-BERT (SBERT) embeddings) with structural awareness (via Graph Neural Networks). We introduce a novel feature extraction pipeline that includes visual link positioning parsed from raw HTML. Our experiments benchmark a semantic heuristic, a feature-based Logistic Regression, and two GNN variants (GraphSAGE and GATv2). We demonstrate that a Hybrid GraphSAGE, which fuses learned node embeddings with explicit shortest-path distances, achieves the highest Mean Reciprocal Rank (MRR) of approximately 0.512, outperforming traditional baselines and demonstrating the efficacy of combining neural representation learning with classical graph heuristics. The Hybrid GATv2 achieves an MRR of 0.491, Logistic Regression achieves 0.474, and the heuristic performs the worst, only achieving an MRR of 0.174.

1 Introduction

Navigating information networks is a fundamental task in the digital age. While computational algorithms can easily compute the shortest path between two concepts, human navigation is bounded by cognitive constraints and driven by "information scent", a mixture of semantic intuition and visual cues. This research addresses the problem of simulating human-like navigation strategies on Wikipedia. Humans rarely follow the theoretically optimal shortest path; instead, they rely on local semantic decisions and background knowledge to traverse the network. Understanding this divergence between algorithmic optimality and human behavior is crucial to improve recommendation systems, optimizing website structures, and designing educational tools that guide users effectively. By bridging the gap between classical graph-based algorithms and cognitive models of pathfinding, we explicitly model features such as "hub-seeking" bias and visual link prominence.

The primary objective of this project is to predict the *next link* a human player will click in the *Wikispeedia* game, given their current article and a target goal. We aim to move beyond simple frequency-based predictions by leveraging modern Natural Language Processing. Specifically, we use pretrained SBERT embeddings to quantify the semantic relevance of candidate links, validating whether deep semantic understanding can accurately forecast human navigation choices.

Our work focuses on the supervised ranking of hyperlinks using the *Wikispeedia* dataset. The scope includes rigorous feature engineering to extract novel signals such as HTML link position, semantic similarity, and graph topology metrics such as PageRank and out-degree. We evaluate and compare three distinct approaches: a rule-based semantic heuristic, a feature-rich Logistic Regression baseline, and a Hybrid Graph Neural Network (GNN) architecture designed to learn complex, non-linear navigation policies.

2 Related Work

The *Wikispeedia* dataset was introduced in 2009 [1]. The dataset was designed to study human navigation behavior within the Wikipedia network by having participants play a game that mimics the process of navigating from one Wikipedia article to another. The paper demonstrated that human navigation does not follow the shortest-path algorithmic approach commonly used in graph theory. Instead, humans tend to rely on semantic intuition and background knowledge when making navigation decisions, often choosing links that seem contextually relevant even if they do not immediately lead to the goal article. The authors showed that shortest-path distance alone was insufficient for capturing human-like behavior in navigating the Wikipedia graph. They found that click frequencies could be a valuable indicator of human navigation patterns, encoding the human preference for semantically closer links. However, this approach did not explicitly model the semantic content of the articles themselves.

The paper titled *The Wikispeedia Game: Are We Playing It All Wrong?* investigates human navigation patterns within the Wikipedia network through the analysis of over 75,000 gameplay paths from the *Wikispeedia* game [2]. The research found that players typically take paths that are 2.4 times longer than the optimal route, indicating significant inefficiencies in their navigation strategies. The authors explored various strategies, including hub-based navigation, semantic-guided choices, and backtracking behaviors. One of the key findings is that human players often rely on "hub"-articles, which are highly connected and central topics within the Wikipedia network, such as "United States" or "Europe." These hubs act as natural stepping stones for navigating the network. However, while using hubs can be effective, the paper emphasizes that this strategy alone is suboptimal and works best when combined with other strategies, such as semantic navigation, where players select links that are semantically closer to their target article. Another key observation is that backtracking, or revisiting previously visited articles, significantly increases the completion time, and successful navigation paths tend to minimize this behavior. The study also highlights the importance of link position, with players often clicking on top-positioned links, suggesting that quick scanning and using prominent links are time-efficient strategies.

3 Dataset, Preprocessing, and Feature Engineering

Dataset:

We utilize the *Wikispeedia* dataset[1], which consists of 4,604 articles and nearly 120,000 links. The data set includes both HTML and plain text versions of each article. In addition, it contains several tabular data files, such as `articles.tsv`, which maps the name of the articles to unique identifiers, `categories.tsv` , which provides hierarchical category information for the articles, and `links.tsv` , which describes the relationships between articles by listing the source and target articles. Additionally, the dataset contains `finished_paths.tsv` and `unfinished_paths.tsv`. These files include information on the navigation paths taken by players, including the start and goal articles, the sequence of articles visited, and the duration of each game. The `finished_paths.tsv` file also includes a difficulty rating for each game, while the `unfinished_paths.tsv` file specifies the reason for the game's termination. Lastly, the `shortest-path-distance-matrix.txt` provides the shortest path distances between each pair of articles in the dataset.

Data Preprocessing:

In the preprocessing step, we extended the articles dataframe by incorporating both the plain text and HTML content corresponding to each article. This process ensures that we have all the textual information needed for feature extraction. A descriptive analysis of the finished paths reveals that the

[1] `https://snap.stanford.edu/data/wikispeedia.html`

average length of a completed game is 6.75 steps, with the shortest game consisting of only 1 step and the longest game containing 435 steps.

Feature Engineering:

After preprocessing, we proceed with feature engineering. First, we calculate and add the length of each article to the dataframe. Additionally, we compute the in-degree and out-degree for each article, which represent the number of other articles linking to the respective article (in-degree) and the number of articles the article links to (out-degree). These features provide insight into the structural properties of each article within the Wikipedia graph. The `links` dataframe is further enhanced by adding the exact position of each link within the article and the relative position of the link (scaled between 0 and 1), where 0 indicates the link is at the beginning of the article and 1 indicates it is at the end.

Moreover, we incorporate graph-based features, including the PageRank of each article. The directed graph constructed from the dataset is represented by articles as nodes and hyperlinks between them as edges. The PageRank algorithm is used to measure the relative importance of each article within the network, with articles that have more incoming links being considered more important.

For semantic feature extraction, we utilize the SBERT model to generate embeddings for each article. These embeddings capture the semantic meaning of the articles to enable the model to understand the content of the articles beyond their structural relationships. To quantify the semantic similarity between articles, we calculate the cosine similarity between their embeddings. For each article in a given step, we compute two key similarity features:

- The similarity between the current article and the candidate article (`sim_source_candidate`)
- The similarity between the candidate article and the goal article (`sim_candidate_goal`)

These cosine similarities serve as crucial features for predicting which link a human is likely to click next, as humans tend to click on links that are semantically closer to their intended goal.

4 Methods

We formulate the problem as a ranking task: given a current article s and a target g, rank all neighbors $c \in \mathcal{N}(s)$ by the probability of being clicked.

4.1 Baselines

The baseline is build on logistic regression; a supervised linear model trained on the concatenated feature vector described. It serves as a strong baseline because of its access to the shortest path feature.

4.2 Heuristic Approach

To predict the next link a human would click, we employ a heuristic that combines two critical factors: semantic similarity and hub-seeking behavior. This heuristic is designed to model human navigation patterns, where players often start by exploring broad, highly connected topics (hub-seeking) and later focus on articles semantically closer to their goal.

The heuristic score for a candidate article c is computed using the following formula:

$$\text{Score}(c) = \alpha \cdot \cos(\mathbf{v}_c, \mathbf{v}_g) \;+\; (1-\alpha) \cdot \frac{\text{outdeg}(c)}{\max_{c'} \text{outdeg}(c')}$$

Where:

- c = candidate next article
- g = goal article
- $\mathbf{v}_c$ = embedding vector of the candidate article

- $\mathbf{v}_g$ = embedding vector of the goal article
- $\cos(\mathbf{v}_c, \mathbf{v}_g)$ = cosine similarity between the embeddings of the candidate and goal articles
- outdeg(c) = number of outgoing links from the candidate article (indicating its hub status)
- $\max_{c'}$ outdeg(c') = maximum out-degree among all candidates (used to normalize the out-degree value)
- $\alpha \in [0, 1]$ = a hyperparameter controlling the trade-off between semantic similarity and hub-seeking behavior. The value of α changes dynamically based on the stage of the game: Early in the game: A larger value of α prioritizes hub-seeking behavior, as players typically explore broad topics with higher out-degree. Later in the game: A smaller value of α emphasizes semantic similarity directing players to articles closer in meaning to their goal.

This heuristic functions by balancing two factors:

1. **Semantic Similarity:** As the game progresses, players are more likely to click on articles that are semantically similar to their goal. This is captured through cosine similarity between the embeddings of the current article and the goal article.
2. **Hub-Seeking Behavior:** Early in the game, players often explore highly connected articles (hubs), which are likely to have many outgoing links. This is captured by the out-degree feature of each candidate article.

The parameter α allows the model to adapt to different phases of the navigation process: exploration at the start of the game (higher α) and concentration on the goal at the later stages (lower α).

This heuristic provides a simple but effective approach to modeling human navigation behavior. It will be evaluated against other models, including logistic regression, to determine its effectiveness in predicting human-like link selection.

4.3 Hybrid Graph Neural Networks

We propose a Hybrid GNN to capture non-linear dependencies between semantics and structure. Unlike standard link prediction which often predicts edge existence, we predict edge selection.

Encoder (GraphSAGE & GATv2): We use a 2-layer GNN to transform initial SBERT embeddings $\mathbf{x}_i$ into context-aware embeddings $\mathbf{h}_i$. We compare two convolution operators:

1. **GraphSAGE:** Aggregates neighbor information using mean pooling.
2. **GATv2:** Uses attention mechanisms to weigh neighbors dynamically, simulating human selective attention.

$$\mathbf{h}_i = \sigma\left(\mathbf{W} \cdot \text{AGG}\left(\{\mathbf{x}_i\} \cup \{\mathbf{x}_j, \forall j \in \mathcal{N}(i)\}\right)\right)$$

Hybrid Predictor (The "Injection" Strategy): A standard GNN often struggles to learn global graph distances (such as the shortest path to the target article) solely from local message passing. To address this, we explicitly inject the scalar distance $d(c, g)$ into the final readout layer. The scoring function is a Multi-Layer Perceptron (MLP):

$$\text{Score}(c) = \text{MLP}\left(\mathbf{h}_s \parallel \mathbf{h}_c \parallel \mathbf{h}_g \parallel d(c, g)\right)$$

This allows the model to leverage deep semantic reasoning ($\mathbf{h}$) while retaining the strong signal of graph proximity (d).

5 Experiments and Results

For all experiments, we used only games with finished paths and applied an 80/20 train–validation split. Training was performed with negative sampling, using one positive example and 5–20 negative examples per step, and optimizing a Binary Cross Entropy loss. We employed the Adam optimizer with a learning rate of 10^{-3} and used gradient accumulation to emulate large batch sizes (effective batch size of approximately 8192) despite limited hardware resources.

We quantitatively evaluate model performance using MRR, with the results reported in Table 1. Mean Reciprocal Rank (MRR) is a ranking-based evaluation metric that computes, over a set of queries, the average of the reciprocal ranks of the first correctly predicted (relevant) item in each ranked list. As illustrated in Figure 1, the GNN-based models consistently outperform all baselines over the course of training, achieving higher MRR scores across epochs.

Table 1: Model Comparison of MRR scores. The Hybrid GNNs outperform both the heuristic and the feature-engineered logistic regression. GraphSAGE achieves the highest performance.

Model	MRR (Best)
Heuristic (Semantic + Hubs)	0.174
Logistic Regression (PoC)	0.474
Hybrid GATv2	0.491
Hybrid GraphSAGE	**0.512**

We observe that the heuristic baseline performs comparatively poorly (MRR $= 0.174$), which is expected given that it lacks an explicit notion of graph distance. In contrast, the logistic regression model achieves a substantially higher score (MRR $= 0.474$), benefitting from manually engineered features such as shortest-path distance and link position. Both GNN architectures outperform this baseline, with Hybrid GraphSAGE achieving the best result (MRR $= 0.512$), indicating that the incorporated attention mechanism effectively weights neighboring nodes and helps to filter out noise in the link prediction task.

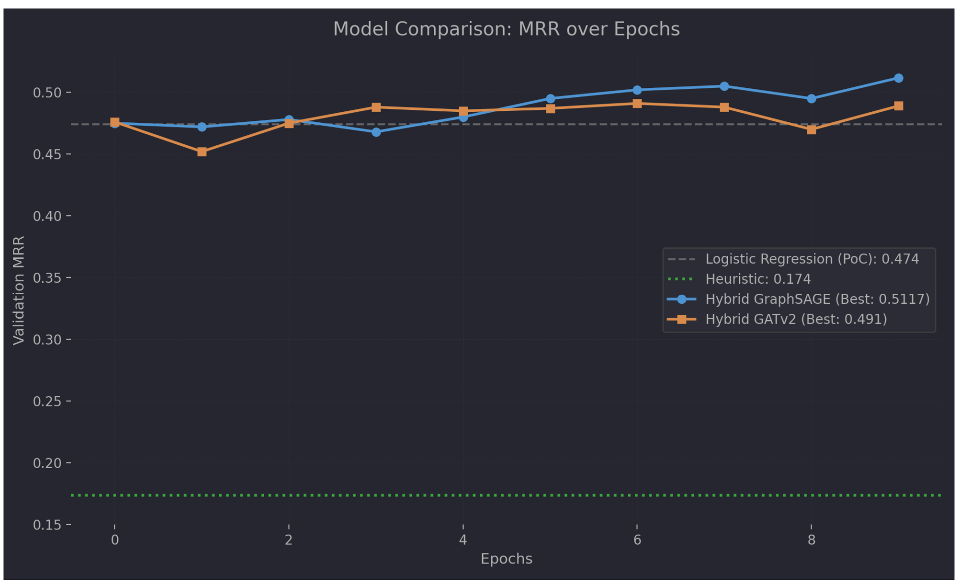


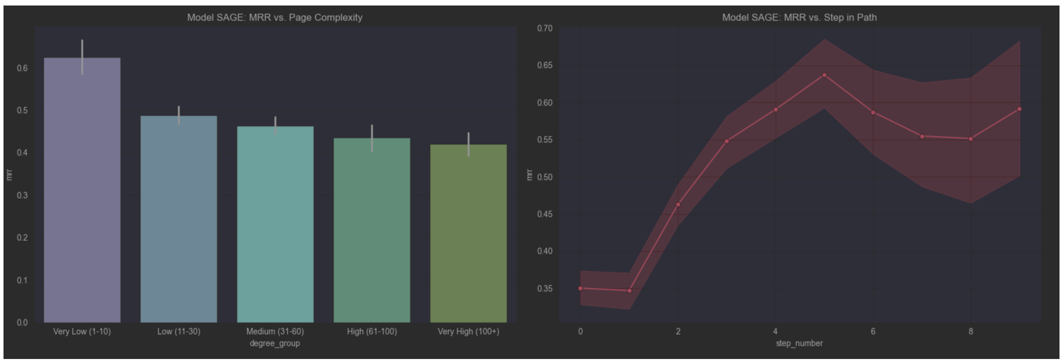
Figure 1: Validation MRR over epochs. Both GNN models (solid lines) converge to a higher MRR than the strong Logistic Regression baseline (grey dashed line).

6 Error Analysis and Discussion

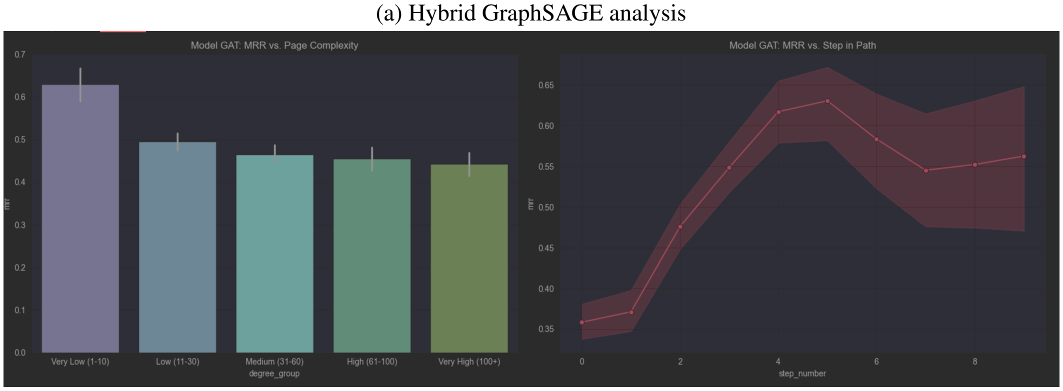
We analyze the performance of the model as a function of page complexity and navigation progress in order to identify the systematic strengths and failure modes of the proposed approaches. Specifically, we report MRR with respect to node out-degree and step number for both Hybrid GraphSAGE and Hybrid GATv2 models, as shown in Figure 2.

Performance decreases monotonically with increasing page complexity. For pages with very low out-degree, both models achieve MRR scores of approximately 0.62, whereas pages with very high out-degree (above 100 outgoing links) exhibit substantially lower performance, with an MRR score in the range of 0.42–0.45. This degradation is expected, as higher out-degree introduces increased candidate ambiguity and dilutes the semantic signal associated with each outgoing edge. Across all complexity regimes, GATv2 consistently matches or slightly outperforms GraphSAGE, with the largest relative gains observed on high-degree nodes. This suggests that the attention mechanism provides improved robustness in dense neighborhoods by selectively weighting semantically informative links.

We further analyze model performance as a function of navigation progress and observe a clear inverted-U relationship with the step index. In the early stages of navigation (steps 0–2), MRR remains relatively low (around 0.35), reflecting the high uncertainty and weak semantic alignment when the target is still several hops away. Performance increases substantially in the mid-navigation phase (steps 3–6), where the models reach peak MRR values above 0.60. At this point, the target concept is typically semantically closer, and contextual text representations become more informative for discriminating between candidate links. In the later stages (steps 7 and beyond), performance stabilizes or slightly declines, which we attribute to path-specific effects such as local oscillations or redundant transitions in the vicinity of the target.



(a) Hybrid GraphSAGE analysis



(b) Hybrid GATv2 analysis

Figure 2: Error analysis for GraphSAGE (a) and GATv2 (b). Left: MRR as a function of page out-degree. Right: MRR as a function of navigation step.

Finally, we assess the contribution of HTML-based structural features using a Logistic Regression model. The link position feature exhibits a statistically significant negative coefficient, indicating a strong preference for links appearing earlier in the page layout. This confirms the presence of a positional bias in human navigation behavior. Incorporating such structural signals into the GNN models enables them to partially account for this bias, contributing to improved ranking performance.

7 Conclusion

We presented a GNN-based approach to modeling human navigation on the *Wikispeedia* dataset that processes HTML structure and combines semantic and structural signals, outperforming strong baselines. Our results suggest that navigation is best captured by a learned integration of both modalities, explicitly conditioned on goal proximity. Future work could explore hard negative mining, incorporate edge attributes (e.g., link position) into message passing, and extend the experimental setup beyond current hardware constraints.

References

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